PRANAV PITHADIA
160104241101
B3

# TUTORIAL 7

Date: 1 / 4 / 26

Q1) (7,4) cyclic code
$g(x) = 1 + x + x^3$

(a) Generator Matrix in non-systematic form

$g = [1\,1\,0\,1\,0\,0\,0]$

$$G = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

(b) To convert to systematic form do row transformations

$R_1 \rightarrow R_1 + R_4$

$R_3 \rightarrow R_3 + R_4$

$R_2 \rightarrow R_2 + R_3$

$R_1 \rightarrow R_1 + R_2$

$$\therefore \quad G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

PTO ⇒

Date: / /

(c) Parity Check matrix of non systematic version of cyclic code

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad \therefore P^T = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

$$H = \begin{bmatrix} P^T & I \end{bmatrix}$$

$$\therefore H = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(d) Codewords $\quad C = m \times G_{sys}$

(i) (1001)

$$C = [1001] \times \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= [1001011]$$

(ii) (1100)

$$C = [1100] \times G$$

$$= [1100101]$$

(iii) (0101)

$$C = [0101] \times G$$

PTO ⇒

$$= [0101110]$$

Date: / /

(iv) (1010)

$C = [1010] \times G$

$= [1010001]$

Q2) Given $g(x) = x^3 + x + 1$

$m = 1010$

$\therefore m(x) = x^3 + x$

$c(x) = m(x) \cdot g(x)$

$= (x^3 + x)(x^3 + x + 1)$

$= x^6 + x^4 + x^3 + x^4 + x^2 + x$

$\therefore c(x) = x^6 + x^3 + x^2 + x$

Converting to binary

$\therefore C = 1001110$

Message is not directly present

$\therefore$ code is non - systematic

PTO $\Rightarrow$

Date: / /

Q3) Given

message = 1010

generator = 1011

$\therefore g(x) = x^3 + x + 1$

$m(x) = x^3 + x$

$c(x) = m(x) \cdot g(x)$

$= (x^3 + x)(x^3 + x + 1)$

$= x^6 + x^4 + x^3 + x^4 + x^2 + x$

$\therefore c(x) = x^6 + x^3 + x^2 + x$

Converting to binary

$c = 1001110$

Message is not directly present

$\therefore$ Non systematic